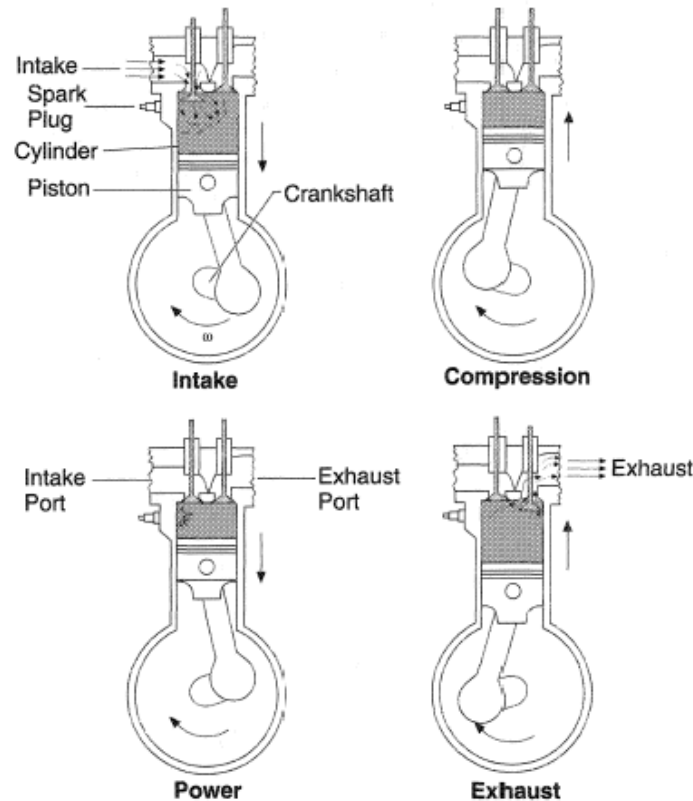


Internal Combustion Engine Analysis

Overview:

An internal combustion engine converts the chemical energy of fuel into mechanical work by mixing fuel and air, burning the mixture inside the engine cylinder and then using the high-pressure combustion gases to push the piston that converts its linear motion into rotational motion via a crankshaft.



Key features:

- Ignition for specifically spark ignition (SI) internal combustion engines comes from the spark plug
- Combustion occurs inside the working fluid (unlike steam or gas turbines with external combustion)
- Operates cyclically
- It can be treated as a closed system during compression/expansion (but not for intake/exhaust)

Components:

- Cylinder - chamber where combustion occurs
- Piston - converts gas pressure to mechanical motion
- Crankshaft - converts reciprocating motion to rotation
- Intake port - admits air/fuel mixture
- Exhaust port - releases combustion products
- Spark plug - initiates combustion

The 4-Stroke Cycle:

1. Intake:
 - The intake valve opens and the piston moves downward as air-fuel mixture enters the cylinder
2. Compression:
 - While both valves are closed, the piston moves upward and the mixture is compressed (pressure and temperature increase)
3. Power (Combustion):
 - Spark ignites mixture and rapid combustion increases pressure. This forces the piston downward and creates work output
4. Exhaust:
 - The exhaust valve opens and the piston moves upward. Combustion products are expelled (mostly carbon dioxide and water vapor)

Balances:

- Mass Balance:

$$m_{air} + m_{fuel} = m_{exhaust}$$
$$\Delta m = 0$$

- Energy Balance:

$$dE_{cv}/dt = Q - W + m_{in} h_{in} - m_{out} h_{out}$$
$$0 = Q_{in} - Q_{out} + W_{shaft} + m_{in} h_{in} - m_{out} h_{out}$$

- Entropy Balance:

$$dS_{cv}/dt = Q/T - W + m_{in} S_{in} - m_{out} S_{out} + S_{gen}$$
$$0 = Q/T - W + m_{in} S_{in} - m_{out} S_{out} + S_{gen}$$

Efficiency:

- Thermal Efficiency:

$$\eta_{th} = W_{out}/Q_{in}$$

This will always be less than one due to heat loss from radiation, and work loss from friction

- Otto Cycle Efficiency (Idealized Model):

$$\eta_{otto} = 1 - 1/r^{\gamma}(\gamma - 1)$$

$r = V_{max}/V_{min}$ (compression ratio)

$$\gamma = c_p/c_v$$

This models the efficiency of the engine if it were both adiabatic and isentropic, neither of which are true for a real engine but we can use this equation to see how we may be able to increase the efficiency

Changes to Increase Efficiency:

- Increase Compression Ratio (r):
 - From the efficiency equation for the Otto Cycle we can see that η_{otto} increases as r increases, thus increasing the ratio V_{max}/V_{min} will result in greater efficiency
- Insulate Engine:

- Insulating the engine would lower the Q_{out} which allows more of the heat input to be converted to work rather than being lost through radiation out of the system
- Brings the system closer to its adiabatic efficiency

Conclusion:

The internal combustion engine operates in four main strokes: intake, compression, power, and exhaust. Although this cycle can be modelled by the idealized Otto Cycle, this idealization neglects the real world irreversibilities of friction and heat transfer across finite temperature differences. Nonetheless, the same factors that would increase the efficiency of the Otto Cycle also improve that of a real IC engine making it a useful comparison.